Ipsita Umang

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SKILLS

- **Languages**: C, C++, Python, Java, JavaScript, HTML5, CSS, SQL
- **Frameworks**: Flask, Django, TensorFlow, PyTorch
- **Libraries**: Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn
- **Tools**: Git, GitHub, Google Colab, Linux, Windows, MS Excel
- **Web Technologies:** REST APIs, JSON, Node.js
- **Databases:** MySQL, DBMS
- **Specialized Skills:** Concurrency, Multithreading, Synchronization, Model Deployment, Data Visualization
- **Soft Skills**: Problem-Solving, Adaptability, Time Management

PROJECTS

Low-Latency Limit Order Book & Matching Engine | Link Jan'2026-Feb'2026

- Engineered a high-performance in-memory exchange core implementing price-time priority matching, enabling deterministic order execution with microsecond-level latency.
- Designed cache-efficient data structures for bid/ask management with O(1) best price access and O(log n) order insertion; sustained ~1.17M+ orders/sec throughput with p50 latency of 600ns and p99 under 1.8µs.
- Integrated an embedded HTTP server and lightweight UI for real-time order submission and trade monitoring to validate matching correctness and performance.
- Tech: C++, STL, CMake, MSVC, HTTP, JSON, Performance Benchmarking

Dropout Prediction System | Link Nov'2025 – Dec'2025

- Created a machine learning–driven dropout prediction solution to detect students at risk in both conventional and online learning settings, facilitating timely intervention and retention strategies.
- Implemented a complete ML workflow including data cleaning, feature extraction, exploratory analysis, and model development using Random Forest and XGBoost; attained a peak accuracy of 75% with the Random Forest algorithm.
- Delivered a scalable, analytics-powered student risk evaluation framework by integrating academic metrics with engagement factors, producing interpretable outcomes applicable to educational institutions.
- Tech: Python, Pandas, Scikit-learn, XGBoost, Random Forest, Matplotlib, Seaborn

CERTIFICATES

- Machine Learning and NLP Basics –Coursera | Certificate Feb'2026
- Supervised Machine Learning: Regression and Classification – Coursera | Certificate Sept'2025
- LLM Engineering: Master AI, Large Language Models & Agents – Udemy | Certificate Sept'2025
- Computer Communications – Coursera | Certificate Nov'2024

ACHIEVEMENTS

- Achieved LeetCode rating of 1642 Dec'2025
- Achieved CodeChef rating of 1474 (2★) Nov'2025
- Achieved Codeforces rating of 1201 (Pupil) Oct'2025
- Earned Problem Solver - Silver Badge on CodeChef Sept'2024

TRAINING

AI ML For Real World Problem Solving-LPU | Certificate

- Built an AI system capable of analyzing emotional and mental health indicators by leveraging multiple input modalities—text, images, and audio—for more accurate psychological assessment.
- Integrated BERT for text, ViT for images, and Wav2Vec for audio, with preprocessing pipelines using OpenCV and Librosa; implemented multimodal feature fusion to enhance prediction performance, achieving **90% model accuracy**.

EDUCATION

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering with AI & ML; CGPA: 7.79 Aug'2023 – May'2027
- **Budding Buds Sr. Sec. School** Tinsukia, Assam
  Intermediate (PCM); **Percentage: 72.6%** Apr'2021 – Mar'2022
- **Budding Buds Sr. Sec. School** Tinsukia, Assam
  Matriculation; **Percentage: 95.2%** Apr'2019 – Mar'2020